

LAB EXPERIMENT RECORD

Set 10: Survey Responses Analysis

Field	Details
Department	AIML
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
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Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart and a data table in R, and to build an interactive dashboard that combines them.

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

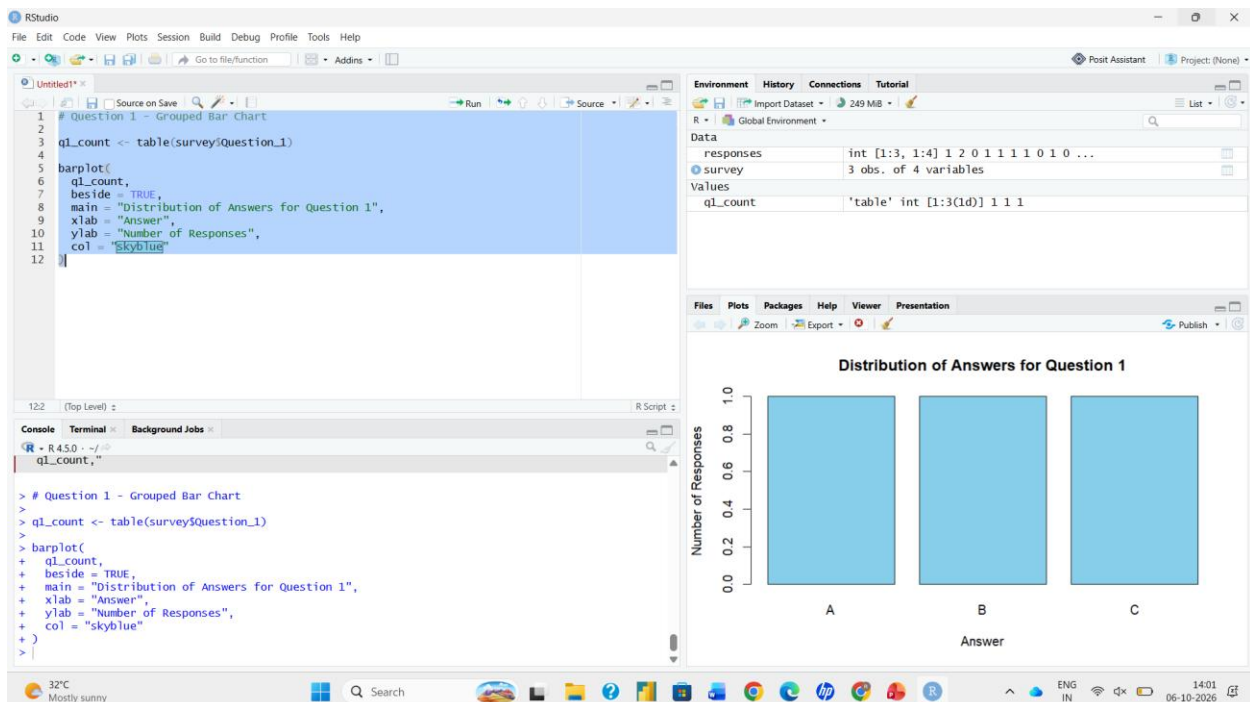
PROGRAM 1: Grouped Bar Chart for Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

# Dataset
survey <- data.frame(
  SurveyID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

# Grouped Bar Chart (Question 1)
ggplot(survey, aes(x = Question1, fill = factor(SurveyID))) +
  geom_bar(position = position_dodge2(preserve = "single")) +
  labs(title = "Distribution of Responses for Question 1",
       x = "Response Option",
       y = "Number of Respondents",
       fill = "Survey ID")
```



Output 1: Question 1 Grouped Bar Chart

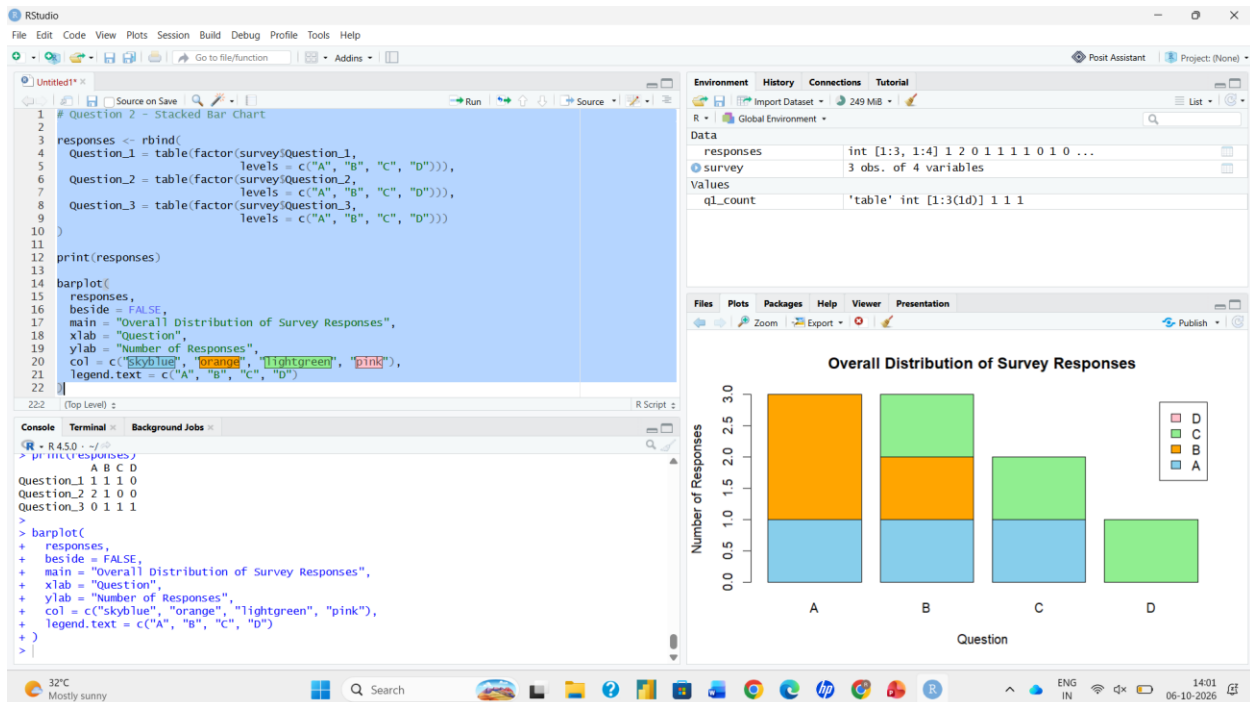
PROGRAM 2: Stacked Bar Chart for All Questions

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)
library(tidyr)

# Convert data to long format
long <- pivot_longer(survey,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response")

# Stacked Bar Chart
ggplot(long, aes(x = Question, fill = Response)) +
  geom_bar(position = "stack") +
  labs(title = "Overall Distribution of Responses (All Questions)",
    x = "Question",
    y = "Number of Responses",
    fill = "Response")
```

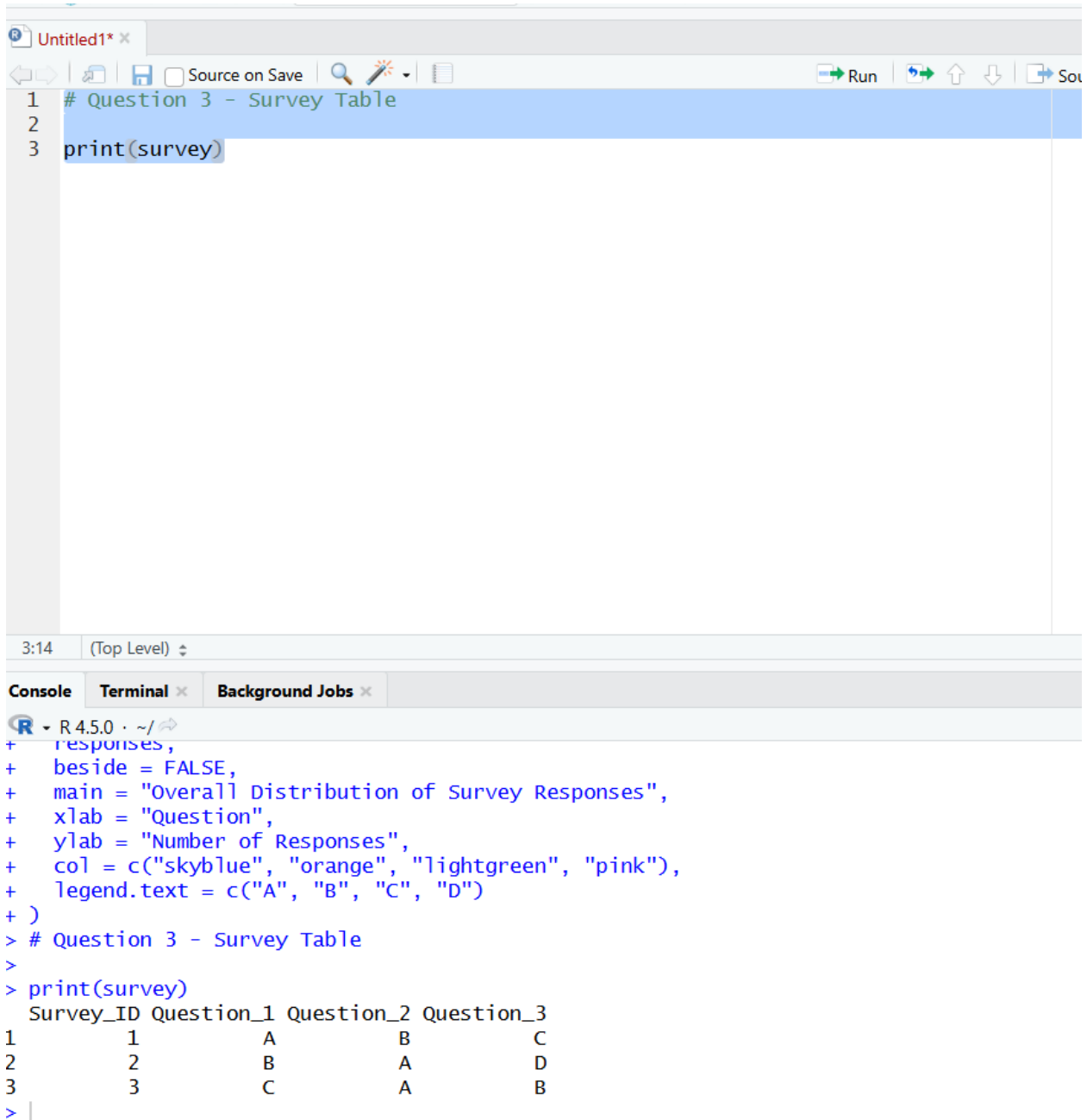


Output 2: Stacked Bar Chart of All Questions

PROGRAM 3: Survey Response Table

Build a table to show the survey response data for each survey. Label the table elements.

```
# Display the survey data as a labelled table
colnames(survey) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
print(survey)
View(survey)
```



The screenshot shows the RStudio IDE interface. The top pane displays the source code for a script named 'Untitled1*.r'. The code defines a data frame 'survey' with four columns: 'Survey ID', 'Question 1', 'Question 2', and 'Question 3'. The data is as follows:

	Survey ID	Question 1	Question 2	Question 3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

The bottom pane shows the console output, which includes the R code being executed and the resulting table. The table is printed with the following structure:

```
> # Question 3 - Survey Table
> 
> print(survey)
  Survey_ID Question_1 Question_2 Question_3
1         1         A         B         C
2         2         B         A         D
3         3         C         A         B
```

Output 3: Survey Response Table

PROGRAM 4: Interactive Shiny Dashboard

Build an interactive dashboard combining the grouped bar chart, stacked bar chart and the table using the Shiny package.

```
library(shiny)
library(ggplot2)
library(tidyr)

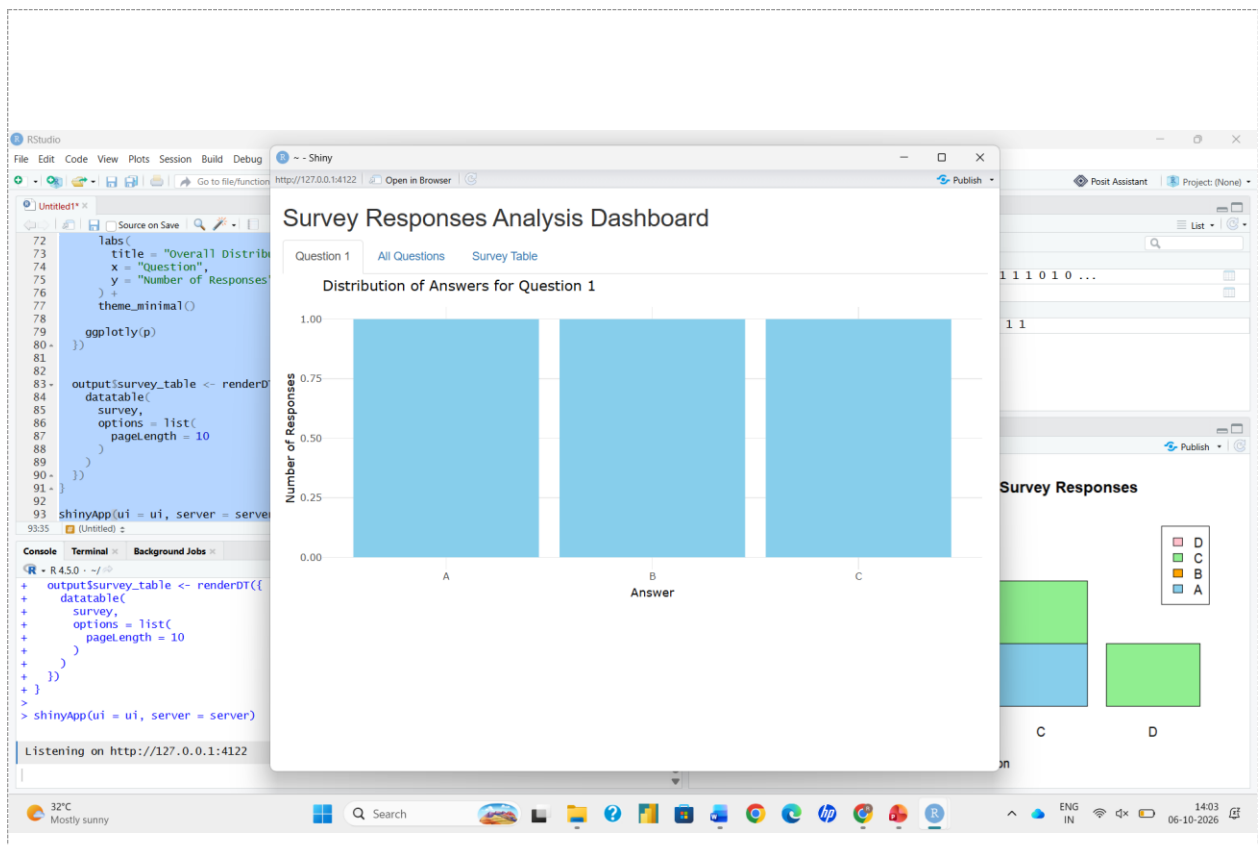
survey <- data.frame(
  SurveyID = c(1, 2, 3),
  Question1 = c("A", "B", "C"),
  Question2 = c("B", "A", "A"),
  Question3 = c("C", "D", "B")
)

long <- pivot_longer(survey, cols = starts_with("Question"),
  names_to = "Question", values_to = "Response")

ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput("q", "Select Question (Grouped Bar Chart):",
        choices = c("Question1", "Question2", "Question3"))
    ),
    mainPanel(
      plotOutput("grouped"),
      plotOutput("stacked"),
      tableOutput("tbl")
    )
  )
)

server <- function(input, output) {
  output$grouped <- renderPlot({
    ggplot(survey, aes(x = .data[[input$q]], fill = factor(SurveyID))) +
      geom_bar(position = position_dodge2(preserve = "single")) +
      labs(title = paste("Distribution of Responses for", input$q),
        x = "Response Option", y = "Number of Respondents",
        fill = "Survey ID")
  })
  output$stacked <- renderPlot({
    ggplot(long, aes(x = Question, fill = Response)) +
      geom_bar(position = "stack") +
      labs(title = "Overall Distribution of Responses (All Questions)",
        x = "Question", y = "Number of Responses", fill = "Response")
  })
  output$tbl <- renderTable({
    colnames(survey) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
    survey
  })
}

shinyApp(ui = ui, server = server)
```



Output 4: Interactive Dashboard